Network Layer: Data Plane Definitions (Ch. 4)

4.1 Network Layer Overview

Datagram The network-layer packet of data.
Forwarding The **local**, per-router action of moving a packet from an input link to the appropriate output link. This is the primary function of the **data plane**.
Routing The **global**, network-wide process of determining the end-to-end path that packets take. This is the primary function of the **control plane**. The output is the forwarding table.
Forwarding Table A table maintained in a router that maps destination addresses (or prefixes) to the router's local output links.
Data Plane The part of the router's architecture that handles the forwarding of datagrams. It operates in hardware on a nanosecond timescale.
Control Plane The part of the router's architecture that is responsible for running routing protocols to compute the forwarding table. It operates in software.
Best-Effort Service The service model of the internet's network layer (IP). It provides **no guarantees** on packet delivery, timing, order, or bandwidth.
SDN (Software-Defined Networking) A network architecture where the control plane is physically separated from the data plane and is logically centralized in a remote controller.

4.2 What's Inside a Router?

Input Port The component of a router that receives an incoming packet. It performs physical layer, link layer, and network layer (lookup) functions at line speed.
Switching Fabric The "network within the router" that connects the input ports to the output ports. Can be implemented with memory, a shared bus, or a crossbar (interconnection network).
Output Port The component of a router that queues packets received from the switching fabric and transmits them over the outgoing link.
Routing Processor The CPU of the router, where the control plane software (routing protocols) runs.
Longest Prefix Matching The rule used by routers to look up a destination IP address in the forwarding table. It selects the table entry with the longest address prefix that matches the destination address.
TCAM (Ternary Content Addressable Memory) Specialized, high-speed memory used in routers to perform longest prefix matching lookups in a single clock cycle.
Head-of-the-Line (HOL) Blocking A performance problem at *input ports*. A packet at the front of a queue is blocked because its output port is busy, preventing all subsequent packets in the queue from moving forward, even if their output ports are free.
Output Port Queuing Buffering that occurs when the switching fabric delivers packets to an output port faster than the output link can transmit them. This is the primary cause of packet loss due to congestion in a router.
Packet Scheduling The discipline used by an output port to decide which packet to transmit next from its queue.
- **FIFO (First-In, First-Out):** Simple, first-come, first-served scheduling.
- **Priority Scheduling:** Always transmits packets from the highest-priority queue that is not empty. Can lead to starvation of low-priority traffic.
- **Round Robin (RR):** Cyclically serves each queue, sending one packet from each in turn.
- **Weighted Fair Queuing (WFQ):** A generalized form of Round Robin where each queue is given a different weight, allowing for differential bandwidth allocation.

4.3 The Internet Protocol (IP)

IP Header The control information at the beginning of an IP datagram. Key fields include:
- **TTL (Time To Live):** A field that is decremented by one at every router. If it reaches 0, the packet is discarded. Prevents infinite routing loops.
- **Protocol Field:** A number that indicates which transport-layer protocol (TCP=6, UDP=17) the payload should be delivered to on the destination host.
- **Fragmentation Fields:** Fields used by a destination host to reassemble an original datagram that was fragmented into smaller pieces by a router along the path because it exceeded a link's **MTU (Maximum Transmission Unit)**.

Interface The connection between a host/router and a physical link. Each interface has its own IP address.
Subnet A group of device interfaces that can reach each other without passing through a router.
CIDR (Classless Inter-Domain Routing) The standard notation for IP addresses and their subnet part, written as 'a.b.c.d/x', where 'x' is the number of bits in the network prefix.
DHCP (Dynamic Host Configuration Protocol) A protocol that allows hosts to automatically obtain an IP address, subnet mask, default gateway, and DNS server address when they join a network.

Route Aggregation The process of advertising a single, large network prefix to the global internet that summarizes multiple smaller, more specific subnets within an ISP's network. This is enabled by hierarchical addressing.
NAT (Network Address Translation) A technique where all devices on a private local network share a single public IP address. The NAT router rewrites the source IP and port number of outgoing datagrams and maintains a **NAT translation table** to reverse the process for incoming replies.
IPv6 The successor to IPv4, featuring a **128-bit address space** and a simplified, fixed-length 40-byte header.
Tunneling A mechanism to transition from IPv4 to IPv6, where an entire IPv6 datagram is encapsulated as the payload within an IPv4 datagram to traverse IPv4-only portions of the internet.

4.4 & 4.5 Generalized Forwarding & Middleboxes

Generalized Forwarding An evolution from simple destination-based forwarding where a router's decision can be based on a combination of header fields from multiple layers (link, network, transport).
Match Plus Action The core paradigm of generalized forwarding. A router's flow table contains rules that "match" on certain header fields and specify an "action" to perform (e.g., forward, drop, modify).
OpenFlow A standard protocol that provides the interface between a centralized SDN controller and the data plane switches. It defines the structure of the **flow table**.
Middlebox A network device that performs functions beyond standard IP forwarding, such as NAT, firewalls, load balancers, and intrusion detection systems. They often operate on Layer 4 or higher information.